

Table 1: Benchmark results for Vanilla, Fast-dLLM Dual Cache, LocalLeap, and BlockBatch across LLaDA-1.5-8B, LLaDA-Instruct-8B, and Dream-Base-7B. For Fast-dLLM Dual Cache, we report only the block size 32 setting. *Notation:* Fast-dLLM denotes Fast-dLLM Dual Cache with block size 32. Green cells indicate BlockBatch results. Subscripts on Latency show speedup over Vanilla; subscripts on NFE show reduction percentage relative to Vanilla.

Task	Method	LLaDA-1.5-8B			LLaDA-Instruct-8B			Dream-Base-7B		
		Lat. (s) ↓	NFE ↓	ACC (%) ↑	Lat. (s) ↓	NFE ↓	ACC (%) ↑	Lat. (s) ↓	NFE ↓	ACC (%) ↑
GSM8K	Vanilla	12.5	256.0	79.91	13.1	256.0	77.10	10.4	256.0	75.13
	Fast-dLLM	4.9 _{↓2.6×}	84.7 _{↓67%}	79.68	2.7 _{↓4.9×}	86.4 _{↓66%}	78.70	3.7 _{↓2.8×}	162.3 _{↓37%}	73.62
	LocalLeap	4.7 _{↓2.7×}	72.5 _{↓72%}	81.27	2.3 _{↓5.7×}	64.4 _{↓75%}	78.16	6.5 _{↓1.6×}	142.9 _{↓44%}	72.71
	BlockBatch	4.2 _{↓3.0×}	63.6 _{↓75%}	77.56	2.4 _{↓5.5×}	63.2 _{↓75%}	77.48	3.7 _{↓2.8×}	133.8 _{↓48%}	72.48
MATH	Vanilla	9.04	256.0	34.78	9.7	256.0	40.14	8.1	256.0	40.10
	Fast-dLLM	5.5 _{↓1.6×}	104.4 _{↓59%}	33.20	3.3 _{↓2.9×}	107.8 _{↓58%}	37.12	2.5 _{↓3.2×}	121.9 _{↓52%}	39.34
	LocalLeap	3.6 _{↓2.5×}	80.3 _{↓69%}	31.74	2.8 _{↓3.5×}	85.3 _{↓67%}	32.18	3.3 _{↓2.5×}	92.8 _{↓64%}	37.98
	BlockBatch	3.4 _{↓2.7×}	77.6 _{↓70%}	32.24	2.4 _{↓4.0×}	80.4 _{↓69%}	37.20	2.2 _{↓3.7×}	92.8 _{↓64%}	39.44
HumanEval	Vanilla	5.77	256.0	43.29	5.7	256.0	40.24	4.7	256.0	50.00
	Fast-dLLM	4.3 _{↓1.3×}	88.7 _{↓65%}	39.02	2.6 _{↓2.2×}	90.4 _{↓65%}	36.59	3.1 _{↓1.5×}	156.1 _{↓39%}	52.44
	LocalLeap	2.0 _{↓2.9×}	75.8 _{↓70%}	42.07	2.1 _{↓2.7×}	69.8 _{↓73%}	35.98	2.6 _{↓1.8×}	123.9 _{↓52%}	48.78
	BlockBatch	2.1 _{↓2.7×}	63.0 _{↓75%}	39.63	2.0 _{↓2.9×}	64.6 _{↓75%}	38.41	2.8 _{↓1.7×}	112.5 _{↓56%}	52.44
MBPP	Vanilla	9.50	256.0	43.40	9.9	256.0	41.40	8.0	256.0	55.80
	Fast-dLLM	4.2 _{↓2.3×}	76.1 _{↓70%}	38.00	2.2 _{↓4.5×}	68.2 _{↓73%}	38.40	2.6 _{↓3.1×}	111.9 _{↓56%}	53.20
	LocalLeap	2.7 _{↓3.5×}	58.0 _{↓77%}	40.20	1.9 _{↓5.2×}	56.1 _{↓78%}	41.40	2.9 _{↓2.8×}	82.8 _{↓68%}	54.60
	BlockBatch	2.6 _{↓3.7×}	54.5 _{↓79%}	40.00	1.7 _{↓5.8×}	45.3 _{↓82%}	39.60	2.3 _{↓3.5×}	81.1 _{↓68%}	52.00